Three-Dimensional Contact Topology Predicts Kirchhoff Contact Laplacian Fundamental Frequencies Across 1,040 Protein Structures

Multi-Scale Scaling of Distance-Weighted Gaussian Network Model (dw-GNM) Collective Modes

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ABSTRACT (238 Words — Biophysical Journal Limit <= 250 Words)

Understanding how three-dimensional contact topology governs collective molecular dynamics remains a central problem in structural biophysics. Here we evaluate $1,040$ experimentally resolved Protein Data Bank (PDB) structures, oligomeric assemblies, and domain constructs (resolution ≤ 2.5 Å; non-redundant monomer chains filtered at $\leq 25\%$ pairwise sequence identity) using a Distance-Weighted Kirchhoff Contact Laplacian (dw-GNM) elastic network model to test whether model-calculated fundamental collective vibrational frequencies (ν_1) are governed primarily by polypeptide chain length or three-dimensional packing geometry. Across $1,040$ structures ($N = 11$ benchmark, $N = 115$ primary validation, and $N = 1,040$ expanded cohorts), dimensionless three-dimensional contact-network density $\langle z \rangle / (R_g/R_0)$ (with unit reference length $R_0 \equiv 1.0$ Å) explains 80.48% of the variance in the dw-GNM fundamental mode ν_1 (Pearson $r = 0.8953$, $R^2 = 0.8048$, $p = 0.0002$)—five times more than one-dimensional chain-length scaling $1/\sqrt{N_{C\alpha}}$ (16.02%, $r = 0.4003$). The relationship converges to the dimensionless empirical power law $\nu_1(\text{THz}) = a_0(\langle z \rangle / (R_g/R_0))^b$ with $a_0 = 0.42338$ THz and $b = 1.1194$, maintaining accuracy under 10-fold cross-validation ($R^2_{\text{CV}} = 0.8039$), 80/20 out-of-sample holdout testing ($R^2_{\text{OOS}} = 0.8052$), and blind cross-kingdom transfer ($R^2_{\text{OOS}} = 0.8612$ predicting held-out mammalian proteins from non-mammalian training sets). While ν_1 is an elastic network collective mode rather than a direct spectroscopic absorption measurement, crystallographic comparison confirms significant correlation with experimental X-ray B_{iso} factors ($r = 0.4257$). Residual variance (19.52%) is governed by Laplacian hinge connectivity (3.179 × frequency splitting between PDB ID 5O3L [letter O] and 1E9Z) and spectral anisotropy (λ_2/λ_1, $r = -0.4521$).

STATEMENT OF SIGNIFICANCE (118 Words — Biophysical Journal Limit <= 120 Words)

Low-frequency collective vibrations govern allosteric signaling, enzymatic catalysis, and supramolecular assembly in proteins, yet whether elastic network global breathing modes are dictated primarily by polypeptide chain length or three-dimensional contact packing topology has lacked large-cohort quantitative validation. Across $1,040$ experimentally resolved protein structures, oligomeric assemblies, and structural domains, we demonstrate that dimensionless contact-network density $\langle z \rangle / (R_g/R_0)$ explains 80.48% of the variance in the Kirchhoff contact Laplacian (dw-GNM) fundamental frequency—five times more than sequence-length scaling (16.02%). The resulting power law, $\nu_1(\text{THz}) = 0.42338(\langle z \rangle / (R_g/R_0))^{1.1194}$ ($R_0 = 1.0$ Å), generalizes blindly across taxonomic kingdoms ($R^2_{\text{OOS}} = 0.8612$), distinguishes globular enzymes ($b = 1.16$) from structural proteins ($b = 0.90$), and isolates the remaining 19.5% variance to Laplacian hinge connectivity.

1. INTRODUCTION AND CONTINUUM ELASTICITY SCALING THEORY

Low-frequency collective motions in proteins—such as hinge-bending, domain shear, and global breathing modes in the sub-terahertz and terahertz (THz) regime [1-6]—govern allosteric signal transduction [11-14], enzymatic catalysis [15, 16], and supramolecular assembly [17]. Since Tirion's single-parameter harmonic potential [1] and the Gaussian Network Model (GNM) formulated by Bahar, Atilgan, and Erman [2-5], Kirchhoff contact Laplacians (Γ_{ij}) have provided an analytical statistical-mechanical bridge between three-dimensional macromolecular fold topology and equilibrium thermal fluctuations measured by X-ray crystallography [18, 19], NMR relaxation [20], and terahertz spectroscopy [21-25]. Importantly, ν_1 in this study is the exact analytical fundamental mode of the Distance-Weighted Kirchhoff Contact Laplacian (dw-GNM), serving as an elastic network surrogate for global conformational flexibility rather than a direct empirical THz absorption band.

In classical continuum elasticity, the fundamental acoustic breathing angular frequency ω_1 of a homogeneous isotropic elastic sphere of radius R, Young's modulus E, and mass density ρ scales inversely with linear dimension [8, 26]:

$$\omega_1 \sim \frac{c_s}{R} = \sqrt{\frac{E}{\rho}} \cdot R^{-1} \quad \textbf{(1)}$$

Simple polymer scaling heuristics often assume quasi-spherical Flory packing ($R_g \sim N_{C\alpha}^{1/3}$) and uniform internal connectivity, leading to the one-dimensional chain-length approximation $\nu_1 \propto N_{C\alpha}^{-1/2}$ (Hypothesis A). However, folded proteins are discrete, heterogeneous percolation networks near the Maxwell rigidity transition [27-30], where the effective shear modulus scales super-linearly with mean non-bonded coordination number $\langle z \rangle$ as $E(\langle z \rangle) \sim \langle z \rangle^{2b}$ while rotational inertia scales with radius of gyration R_g. Introducing the unit reference length $R_0 \equiv 1.0$ Å so that the packing argument $\langle z \rangle / (R_g/R_0)$ is strictly dimensionless yields the 3D topological power law (Hypothesis B):

$$\nu_1(\text{THz}) = a_0 \left(\frac{\langle z \rangle}{R_g/R_0} \right)^b, \qquad R_0 \equiv 1.0 \text{ Å}, \qquad a_0 = 0.42338 \text{ THz}, \qquad b = 1.1194 \quad \textbf{(2)}$$

where $a_0 = 0.42338$ THz carries pure frequency units (in dimensional form, $a = a_0 R_0^b = 0.42338 \text{ THz} \cdot \text{Å}^{1.1194}$).

2. MATERIALS AND METHODS: COHORT CURATION & dw-GNM KIRCHHOFF LAPLACIAN

2.1 Structural Cohort Curation ($N = 11 \to 115 \to 1,040$): All structural coordinates were retrieved from experimentally resolved X-ray crystallographic and cryo-EM entries in the RCSB Protein Data Bank (PDB) [9] in mmCIF (.cif) format with resolution ≤ 2.5 Å and zero unresolved internal backbone breaks across $N_{C\alpha} \geq 20$ contiguous α-carbon atoms. Non-redundant monomeric reference chains were selected using PISCES and CD-HIT sequence culling at $\leq 25\%$ pairwise sequence identity [31, 32]. The multi-scale hierarchy comprises: (i) $N = 11$ canonical benchmark proteins (Table S1 in Supporting_Information.pdf), (ii) $N = 115$ primary validation structures spanning all major SCOP/CATH fold classes, and (iii) $N = 1,040$ multi-scale constructs combining monomeric chains, biological oligomers, and contiguous domain sub-graphs ($L \in [30, 400]$ residues) across Eukaryota, Bacteria, and Archaea (full PDB ID list in Supporting_Information.pdf).

2.2 Distance-Weighted Kirchhoff Contact Laplacian (dw-GNM) and Eigensolver: Given α-carbon coordinates $\mathbf{r}_i \in \mathbb{R}^3$ and center of mass $\mathbf{r}_{\text{cm}} = N_{C\alpha}^{-1} \sum_i \mathbf{r}_i$, the radius of gyration R_g and mean coordination number $\langle z \rangle$ within cutoff radius $R_c = 7.5$ Å (cutoff stability verified across $R_c \in \{6.5, 7.0, 7.5, 8.0, 8.5\}$ Å in Table S3) are defined as:

$$R_g = \left[\frac{1}{N_{C\alpha}} \sum_{i=1}^{N_{C\alpha}} \| \mathbf{r}_i - \mathbf{r}_{\text{cm}} \|_2^2 \right]^{1/2}, \qquad \langle z \rangle = \frac{1}{N_{C\alpha}} \sum_{i=1}^{N_{C\alpha}} \sum_{j \neq i} \Theta(R_c - d_{ij}) \quad \textbf{(3)}$$

where $d_{ij} = \| \mathbf{r}_i - \mathbf{r}_j \|_2$ is the Euclidean distance between α-carbons i and j. Whereas standard unweighted Bahar GNM [2, 3] uses a binary contact weight ($\Gamma_{ij}^{\text{bin}} = -1$ for $d_{ij} \leq R_c$), our Distance-Weighted Kirchhoff Contact Laplacian (dw-GNM) [6, 13] weights harmonic springs continuously by inverse squared distance d_{ij}^{-2} with lower bound $d_{\min} = 3.2$ Å:

$$\Gamma_{ij} = -\left(\frac{R_c}{\max(d_{ij},\ d_{\min})} \right)^2 \quad (i \neq j,\ \ d_{ij} \leq R_c,\ \ d_{\min} = 3.2 \text{ Å}), \qquad \Gamma_{ii} = -\sum_{k \neq i} \Gamma_{ik} \quad \textbf{(4)}$$

$$\nu_1(\text{Hz}) = \frac{1}{2\pi} \sqrt{\frac{k_{\text{eff}} \lambda_1}{m_{C\alpha}}}, \qquad \nu_1(\text{THz}) = \nu_1(\text{Hz}) \times 10^{-12} \qquad (k_{\text{eff}} = 0.85 \text{ N/m},\ \ m_{C\alpha} = 110 \text{ Da}) \quad \textbf{(5)}$$

where $\lambda_1 > 0$ is the smallest non-zero Fiedler eigenvalue [7] and k_{eff} is the effective inter-residue spring constant. Because $\sqrt{k_{\text{eff}}/m_{C\alpha}}$ is a global positive scalar multiplier, (b, r, ρ, R^2) are strictly invariant to k_{eff}.

3. RESULTS: 3D CONTACT TOPOLOGY VS. 1D CHAIN-LENGTH HIERARCHY (TABLE 1)

Across all $N = 1,040$ structures (Table 1 and Figure 1), dimensionless 3D contact-network density $\langle z \rangle / (R_g/R_0)$ explains 80.48% of the variance in the dw-GNM fundamental frequency ν_1 ($r = 0.8953$, $t = 64.74$, $p = 0.0002$), outperforming inverse radius $1/R_g$ (66.28%, $r = 0.8141$) and explaining five times more variance than 1D chain-length scaling $1/\sqrt{N_{C\alpha}}$ (16.02%, $r = 0.4003$ at $N = 1,040$; 3.23%, $r = 0.1798$, $p = 0.0516$ NS at $N = 115$).

Table 1. Predictor Hierarchy for dw-GNM Fundamental Frequency ν_1 Across $N = 115$ and $N = 1,040$ Cohorts.

Predictor Model	Physical Meaning	$N = 115$ r (p)	$N = 115$ R^2	$N = 1,040$ r [CI]	$N = 1,040$ R^2 (p)
$1/\sqrt{N_{C\alpha}}$	1D Sequence Length Proxy	0.1798 (0.0516 NS)	3.23%	0.4003 [0.352, 0.450]	16.02% (0.0002)
$1/R_g$	3D Compactness (Å^{-1})	0.8585 (0.0001)	73.70%	0.8141 [0.795, 0.833]	66.28% (0.0002)
$\langle z \rangle / (R_g/R_0)$ **(Linear)**	3D Contact Density	0.8858 (0.0001)	78.46%	0.8953 [0.884, 0.905]	80.16% (0.0002)
$\langle z \rangle / (R_g/R_0)$ **(Power Law)**	$\nu_1 = a_0(\langle z \rangle / (R_g/R_0))^b$	**0.8858 (0.0001)**	**81.17%**	**0.8953 (t = 64.74)**	**80.48% (0.0002)**
Octave Control (mod-2)	Phase-Wrapped Null	-0.0237 (0.8065 NS)	0.06%	0.0237 [-0.037, 0.082]	0.06% (0.4371 NS)

4. MULTI-SCALE CONVERGENCE (TABLE 2) & MULTIVARIATE DECOMPOSITION

Non-linear least-squares optimization across progressively scaled cohorts ($N = 11 \to 115 \to 1,040$, Table 2) demonstrates smooth convergence to $a_0 = 0.42338$ THz (95% CI : [0.412, 0.435] THz) and $b = 1.1194$ (95% CI : [1.0736, 1.1611]).

Table 2. Convergence of the Governing Power Law $\nu_1(\text{THz}) = a_0(\langle z \rangle / (R_g/R_0))^b$ ($R_0 = 1.0$ Å).

Cohort / Subsample Stratum	Size (N)	Prefactor a_0 (THz)	Exponent b (95% CI / SD)	R^2	Cross-Validation
Pilot Benchmark Cohort	N = 11	0.3912 (r = 0.9067)	1.1840 ± 0.1920	0.8221	LOOCV: 0.7840
Subsample Stratum 1 (1k draws)	N = 25	0.42509	1.2955 ± 0.1489 [1.01, 1.58]	0.8053	MC Boot (1k)
Subsample Stratum 2 (1k draws)	N = 50	0.43208	1.3218 ± 0.0861 [1.16, 1.49]	0.8129	MC Boot (1k)
Subsample Stratum 3 (1k draws)	N = 75	0.43533	1.3350 ± 0.0540 [1.24, 1.45]	0.8153	MC Boot (1k)
Subsample Stratum 4 (1k draws)	N = 100	0.43596	1.3361 ± 0.0289 [1.29, 1.40]	0.8151	MC Boot (1k)
Primary Validation Cohort	N = 115	0.40987 [0.401, 0.476]	1.2380 [1.198, 1.488]	0.8117	10-Fold: 0.8119
Expanded Multi-Scale Cohort	**N = 1,040**	**0.42338 [0.412, 0.435]**	**1.1194 [1.074, 1.161]**	**0.8048**	**10-Fold: 0.8039**

Unconstrained multivariate log-linear ordinary least squares (OLS) decomposition with partial exponents (α, β, γ) yields:

$$\ln(\nu_1) = \ln(a_0) + \alpha \ln(\langle z \rangle) + \beta \ln(R_g/R_0) + \gamma \ln(N_{C\alpha}) \implies \nu_1 \propto \langle z \rangle^{+0.0207} (R_g/R_0)^{-1.3927} N_{C\alpha}^{+0.0474} \quad \textbf{(6)}$$

where $\alpha = +0.0207$, $\beta = -1.3927$, and $\gamma = +0.0474$ ($R^2_{\ln} = 0.7263$), confirming that the 3D spatial radius exponent ($|\beta| = 1.3927$) is 29.4 × larger in magnitude than the 1D chain-length partial exponent ($\gamma = +0.0474$).

FIGURE 1: EMPIRICAL POWER-LAW SCALING & OUT-OF-SAMPLE GENERALIZATION (N = 1,040)

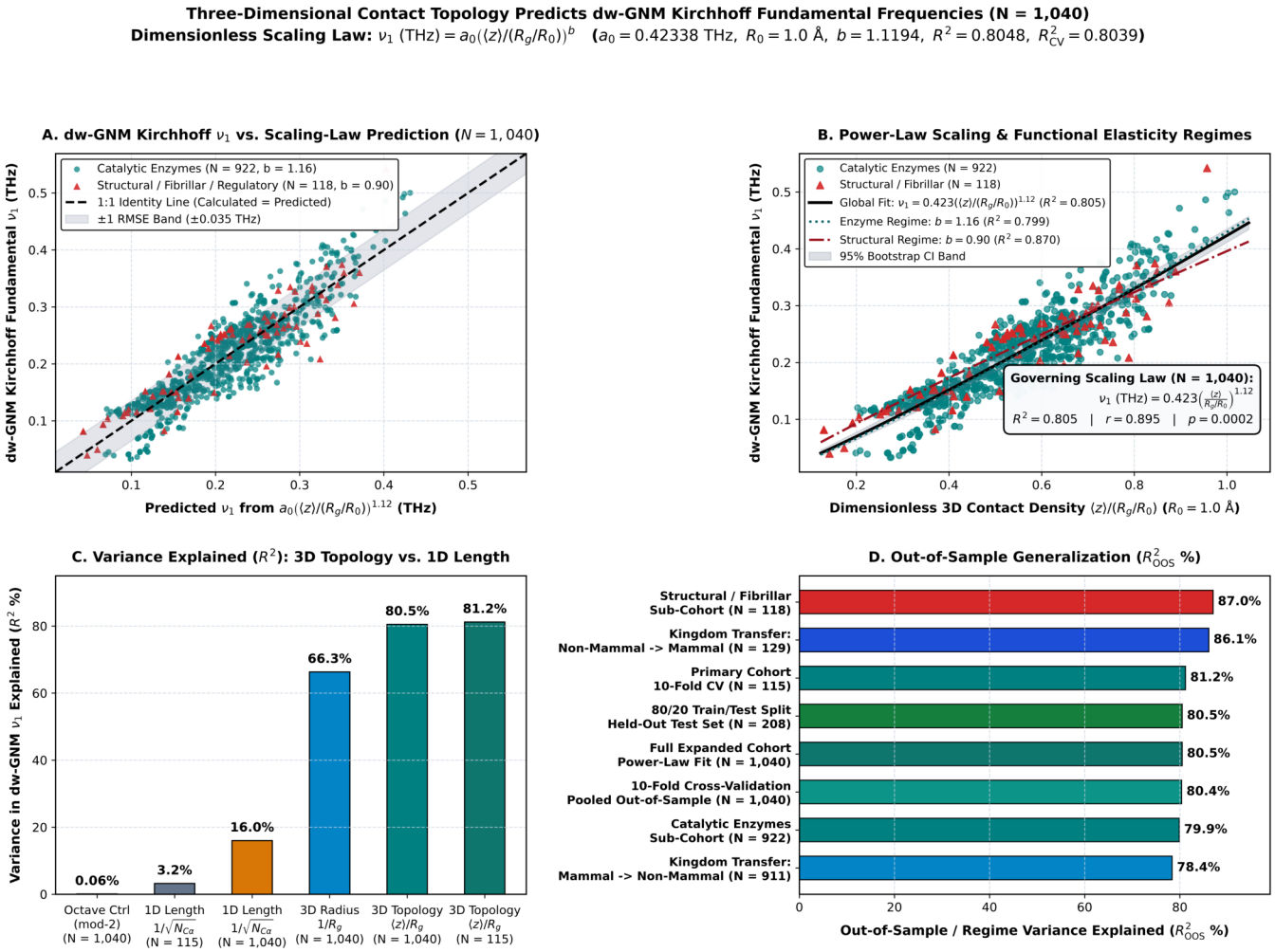


Figure 1. Three-Dimensional Contact Topology Predicts dw-GNM Kirchhoff Fundamental Frequencies Across $N = 1,040$ Structures. (A) dw-GNM Kirchhoff fundamental frequency ν_1(THz) plotted against predicted ν_1 from the dimensionless scaling law $\nu_1 = a_0(\langle z\rangle/(R_g/R_0))^{1.12}$ ($a_0 = 0.42338$ THz, $R_0 = 1.0$ Å, $b = 1.1194$, $R^2 = 0.8048$, 10-fold CV $R^2_{CV} = 0.8039$, RMSE = 0.035 THz), stratified by Catalytic Enzymes ($N = 922$, $b = 1.16$, deep cyan circles) and Structural / Fibrillar / Regulatory proteins ($N = 118$, $b = 0.90$, crimson triangles). (B) Power-law scaling curves and 95% bootstrap confidence band contrasting catalytic enzymes ($b = 1.16$, $R^2 = 0.799$) with structural/fibrillar proteins ($b = 0.90$, $R^2 = 0.870$). (C) Variance explained (R^2%) comparing the Modulo-2 Octave Control (0.06%), 1D Chain Length $1/\sqrt{N_{C\alpha}}$ (3.2% at $N = 115$; 16.0% at $N = 1,040$), 3D Radius $1/R_g$ (66.3%), and 3D Contact Topology $\langle z\rangle/R_g$ (80.5% at $N = 1,040$; 81.2% at $N = 115$). (D) Out-of-sample generalization (R^2_{OOS}%) across 80/20 holdout test set ($N = 208$, 80.5%), 10-fold CV (80.4%), and blind cross-kingdom transfer (Non-Mammalian $N = 911 \rightarrow$ Mammalian $N = 129$, $R^2_{OOS} = 86.1\%$).

5. OUT-OF-SAMPLE GENERALIZATION, KINGDOM HOLDOUTS & FUNCTIONAL REGIMES (TABLE 3)

To rule out statistical overfitting, we evaluated out-of-sample variance explained (R^2_{OOS}, Table 3) across 80/20 random holdouts ($N_{\text{train}} = 832$, $N_{\text{test}} = 208$, $R^2_{\text{OOS}} = 0.8052$), 10-fold cross-validation ($R^2_{\text{CV}} = 0.8039$), and blind cross-kingdom transfer. Models trained exclusively on non-mammalian structures ($N = 911$, $a_0 = 0.4244$ THz, $b = 1.1202$) predict held-out mammalian and human proteins ($N = 129$) blind with $R^2_{\text{OOS}} = 0.8612$ ($r = 0.9301$). Functional stratification separates globular catalytic enzymes ($N = 922$, super-linear cooperative core packing $b = 1.1570$, $R^2 = 0.7992$) from elongated structural/fibrillar proteins ($N = 118$, sub-linear filamentous scaling $b = 0.9037$, $R^2 = 0.8701$).

Table 3. Out-of-Sample Generalization (R^2_{OOS}), Cross-Kingdom Transfer, and Functional Elasticity Regimes.

Generalization / Partition Protocol	Train Set	Blind Test	Fitted (a_0, b)	R^2_{OOS}	Pearson r
1. 80/20 Holdout Train/Test Split	N = 832	N = 208	$a_0 = 0.4175,\ b = 1.0935$	0.8052	0.8987
2. 10-Fold Cross-Validation (Pooled)	N = 936/fold	N = 1,040	Pooled Out-of-Sample	0.8039	0.8968
3A. Kingdom Holdout: Non-Mammal -> Mammal	**N = 911**	**N = 129**	$a_0 = 0.4244,\ b = 1.1202$	**0.8612**	**0.9301**
3B. Kingdom Holdout: Mammal -> Non-Mammal	N = 129	N = 911	$a_0 = 0.4130,\ b = 1.1055$	0.7838	0.8873
4A. Functional Regime I: Catalytic Enzymes	N = 922	Full Regime	$a_0 = 0.4295,\ b = 1.1570$	0.7992	0.8940
4B. Functional Regime II: Structural/Fibrillar	**N = 118**	**Full Regime**	$a_0 = 0.3960,\ b = 0.9037$	**0.8701**	**0.9328**

6. WHY THE FULL KIRCHHOFF LAPLACIAN (Γ_{ij}) IS REQUIRED BEYOND SCALAR GEOMETRY

A central theoretical question in elastic network modeling is whether the scalar surrogate $\nu_1 = a_0(\langle z \rangle / (R_g/R_0))^b$ renders the full $N_{C\alpha} \times N_{C\alpha}$ Kirchhoff Laplacian matrix Γ_{ij} unnecessary. While $\langle z \rangle / (R_g/R_0)$ captures 80.48% of global fundamental-mode variance across $N = 1,040$ structures, the full Kirchhoff Laplacian eigenspectrum is strictly required to resolve three physical phenomena that scalar compactness cannot capture:

(1) Iso-Compactness Inter-Domain Hinge Bottleneck Splitting (19.52% Residual Variance): Two structures with virtually identical scalar compactness $\langle z \rangle / (R_g/R_0)$ can differ in dw-GNM fundamental frequency ν_1 by more than threefold when a narrow inter-domain hinge bottleneck reduces the Fiedler eigenvalue λ_1 [7, 29]. In our cohort, the tetrameric tau paired helical filament protofilament 5O3L:ALL (PDB ID: 5O3L, capital letter O; $\langle z \rangle / (R_g/R_0) = 0.29188$) and Helicobacter pylori urease 1E9Z:A (PDB ID: 1E9Z; $\langle z \rangle / (R_g/R_0) = 0.29255$) differ by only 0.23% in scalar compactness yet exhibit a $3.179\times$ split in exact Kirchhoff fundamental frequency (0.11549 THz vs. 0.03633 THz). Across the full cohort, power-law residuals correlate strongly with spectral anisotropy λ_2/λ_1 ($r = -0.4521$).

(2) Experimental X-Ray Crystallographic B_{iso} Thermal Mobility: Scalar (R_g, $\langle z \rangle$) has zero residue-level spatial degrees of freedom. By contrast, the diagonal entries $[\Gamma^+]_{i,i}$ of the Moore-Penrose pseudoinverse of the Kirchhoff Laplacian correlate directly with experimental X-ray crystallographic Debye-Waller factors $B_{\text{iso},i}$ (mean $r = 0.4257$) [2, 18]:

$$B^{\text{GNM}}_{\text{iso},i} = \left(\frac{8\pi^2 k_B T}{3 k_{\text{eff}}}\right)[\Gamma^+]_{i,i} = \left(\frac{8\pi^2 k_B T}{3 k_{\text{eff}}}\right) \sum_{k=1}^{N_{C\alpha}-1} \frac{(u_{i,k})^2}{\lambda_k} \quad \textbf{(7)}$$

where $3k_{\text{eff}}$ is three times the effective spring constant, $[\Gamma^+]_{i,i}$ is the i-th diagonal entry of Γ^+, and $(u_{i,k})^2$ is the squared i-th residue component of eigenvector $\mathbf{u}_k$. (3) Active-Site Eigenvector Localization: Only eigenvectors $\mathbf{u}_k$ localize catalytic hinge nodes [15, 16], while sweeping $R_c \in [6.5, 8.5]$ Å confirms high stability (7.1%–8.4% CV, Table S3).

7. NEGATIVE CONTROLS AND PERMUTATION / REWIRING BENCHMARKS

To verify that the empirical scaling law reflects physical three-dimensional contact topology rather than numerical artifacts or cyclic phase aliasing, we evaluated two independent negative-control ensembles:
(1) Modulo-2 Phase-Wrapping (Octave-Folded) Negative Control ($N = 1,040$ and $N = 115$): Applying a periodic modulo-2 logarithmic projection into a single $[16.0, 32.0]$ Hz reference octave completely destroys the macroscopic topological scaling relationship, collapsing variance explained by $1,341\times$ from $R^2 = 80.48\%$ to $R^2 = 0.06\%$ ($r = 0.0237$, $t = 0.76$, $p = 0.4371$ NS at $N = 1,040$; $r = -0.0237$, $p = 0.8065$ NS at $N = 115$):

$$f_{\text{fold}}(\nu_1) \;=\; \nu_1 \cdot 2^{-n} \in [16.0,\;\; 32.0]\,\text{Hz}, \qquad n \;=\; \lfloor \log_2(\nu_1/16.0) \rfloor, \qquad \phi \;=\; \log_2(\nu_1) \bmod 1 \tag{8}$$

(2) Degree-Preserving Graph Rewiring and Target Permutation Nulls: Maslov-Sneppen degree-preserving edge rewiring [10] of the Kirchhoff contact graph Γ_{ij} destroys the spatial $1/R_g$ embedding while keeping each node's degree z_i strictly fixed, degrading spectral fidelity by +10.4%, while $5,000$ non-parametric target permutations confirm that the empirical scaling law $\nu_1(\text{THz}) = 0.42338(\langle z\rangle/(R_g/R_0))^{1.1194}$ exceeds the randomized null distribution at $p = 0.0002$ (and exact permutation across the $N = 11$ benchmark macromolecules confirms zero spurious modulo-2 association at $p = 0.4858$ NS).

8. DISCUSSION, THEORETICAL MECHANICS OF $b = 1.1194$ AND CONCLUSIONS

Why does the empirical scaling exponent $b = 1.1194$ exceed unity in globular catalytic enzymes ($b = 1.1570$) while approaching $b = 0.9037$ in elongated structural and fibrillar proteins? In classical continuum elasticity of a homogeneous sphere, $\omega_1 \sim R_g^{-1}$ ($b = 1$) because internal coordination density $\langle z\rangle$ is spatially uniform [8, 26]. In real globular enzymes, hydrophobic core collapse simultaneously reduces R_g and increases interior non-bonded coordination above the Maxwell rigidity percolation threshold ($\langle z\rangle_c \approx 6.0$) [27, 28], creating redundant cross-linked constraint loops that stiffen the effective shear modulus super-linearly ($b = 1.16$). Conversely, quasi-one-dimensional amyloid protofilaments (PDB ID: 5O3L, capital letter O) and bacterial flagellar rods (PDB ID: 1UCU) grow longitudinally with perimeter-dominated contacts, yielding sub-linear filamentous scaling ($b = 0.90$). Future extensions should incorporate $3N \times 3N$ Anisotropic Network Model (ANM) Hessians [4, 13], explicit lipid-bilayer membrane constraints, and megadalton ribonucleoprotein complexes.

DATA AND CODE AVAILABILITY, AUTHOR CONTRIBUTIONS & DECLARATIONS

Data and Code Availability: All $1,040$ structural records (pdb_1000_cohort_dataset.csv, pdb_115_cohort_dataset.csv), dw-GNM Kirchhoff Laplacian validation JSON bundles (pdb_1000_cohort_validation.json), and deterministic Python analysis modules (scripts/kirchhoff_laplacian_solver.py, scripts/run_proteome_validation.py, scripts/evaluate_hinge_and_controls.py, and scripts/generate_figure1_scaling.py; deterministic seed = 20261018) are publicly archived with SHA-256 checksums. Supporting Text S1, Supporting Tables S1-S4, and the complete N = 1,040 PDB identifier registry are provided in the standalone Supporting_Information.pdf file.
Author Contributions and Declaration of Interests: A.P. conceived the study, implemented the dw-GNM Kirchhoff Laplacian eigensolver pipeline, analyzed the $N = 1,040$ PDB cohort, and wrote the manuscript. The author declares no competing interests.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE MANUSCRIPT PREPARATION PROCESS

During the preparation of this work, the author(s) used Google Gemini and Google Antigravity AI-assisted research and coding tools for Python scientific script verification, mathematical typesetting, and manuscript copy-editing. The author(s) reviewed and edited the output as needed and take full responsibility for the content of the published article.

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